Weekly report

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September 21, 2026

Route of the paper

- Study the intermediate rates of forgetting
- Study the competition model

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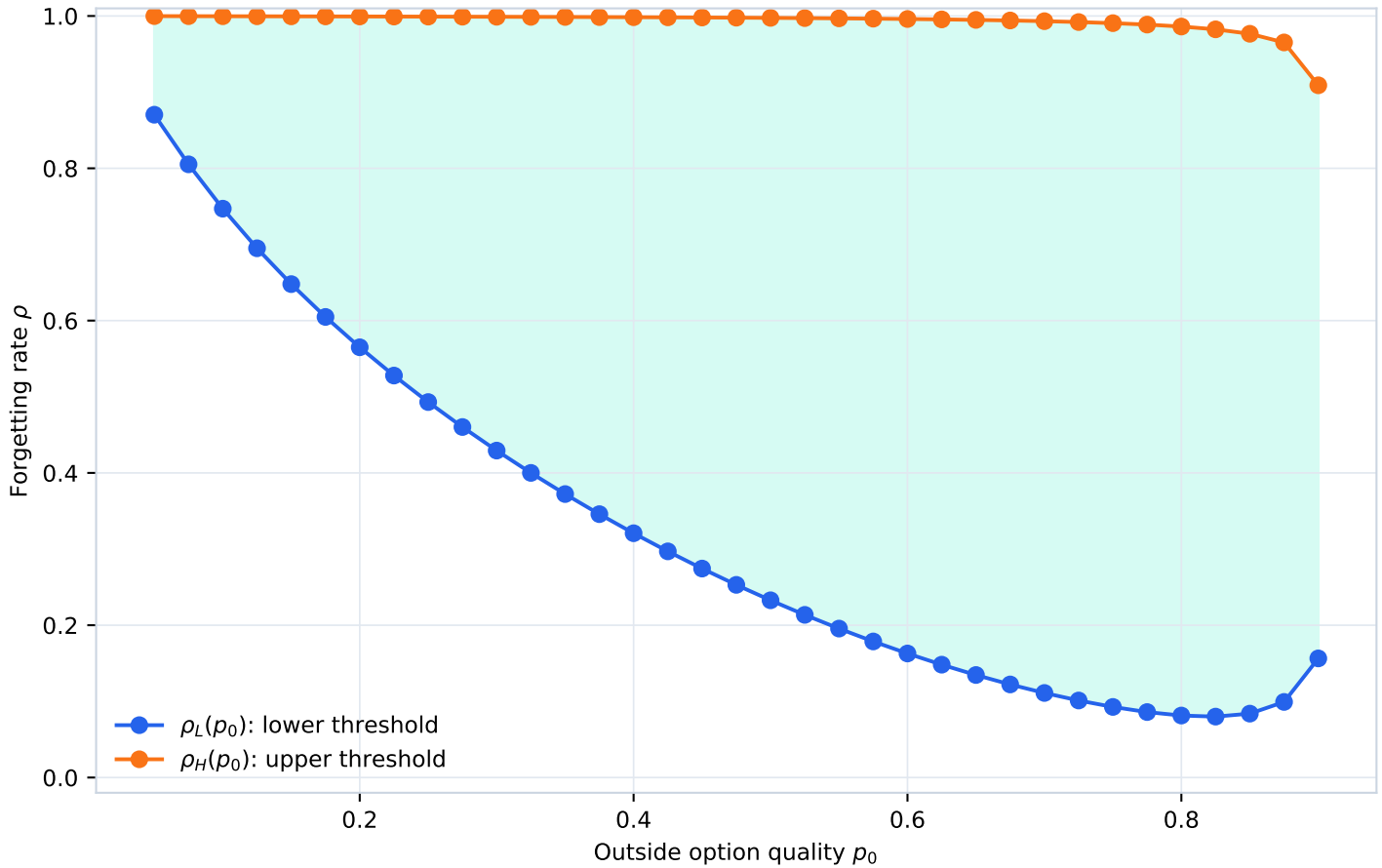
From last meeting

For the forgetting case we have

- Results for the extreme rates (slow and fast forgetting meaning $\rho \to 0$ and $\rho \to 1$)
- Plot of use of product 2 for different values of p_0 and ρ
- Montecarlo simulations

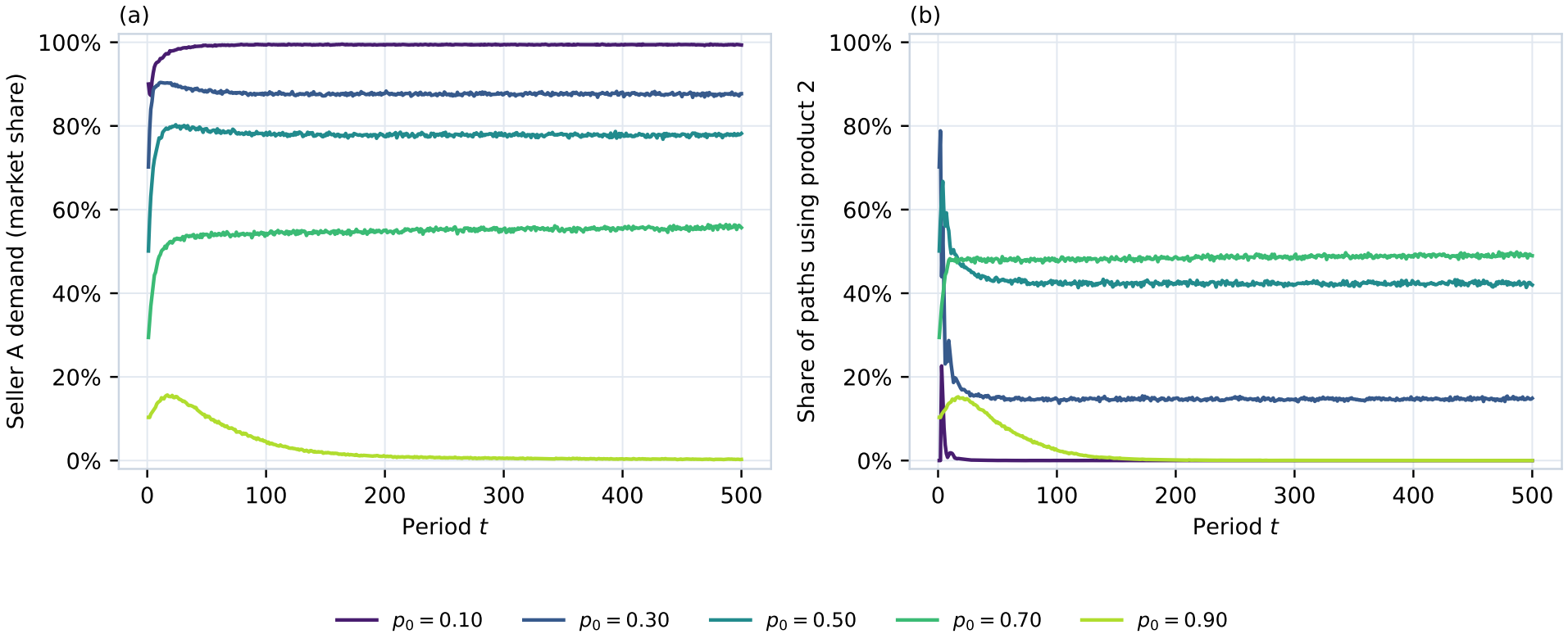
Region of use of product 2

The shaded region only indicates that product 2 is optimal in at least one state for each pair (p_0, ρ).



Monte Carlo simulations: $\rho = 0.95$

Something that we should note is that the montecarlo simulations don't show the frequency of use of product 2 in a single trajectory, but the percentage of trajectories that use product 2 at a certain period of time.



so. . .

- We can plot a single trajectory to see the frequency of use of product 2 over time

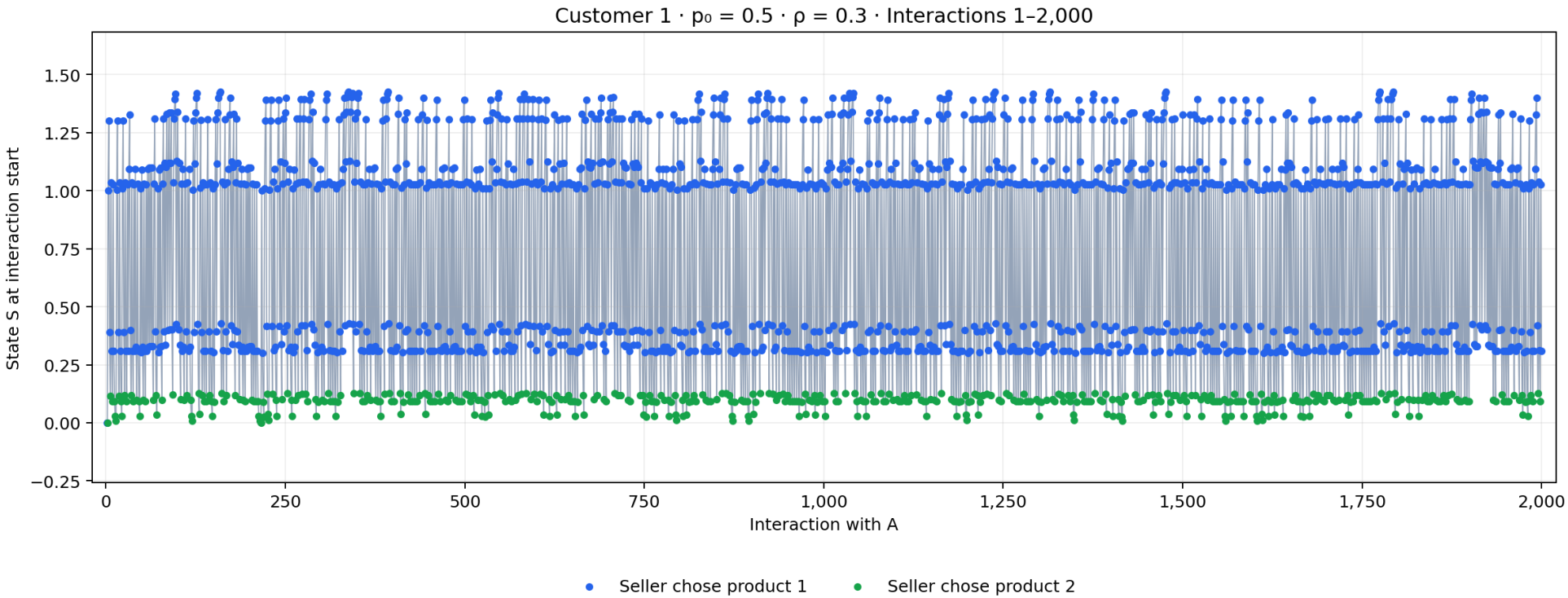
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- When the selection of the provider is not frequent, plotting the trajectory in calendar time produces long periods in which the state does not change.

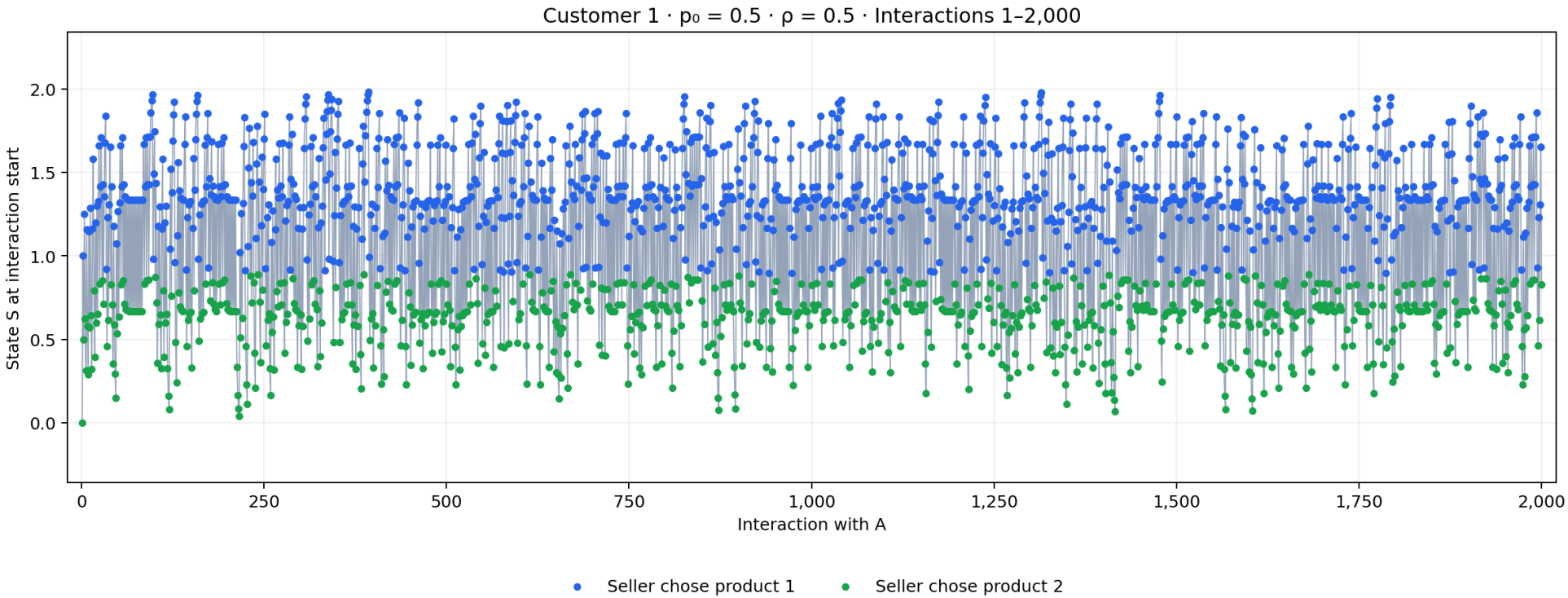
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- We can plot a single trajectory to see the frequency of use of product 2 over time
- When the selection of the provider is not frequent, plotting the trajectory in calendar time produces long periods in which the state does not change.
- We can index the trajectory by interactions with the provider. This only changes the plot's axis, not the solution. The optimal policy is calculated through the bellman equation that already accounts for the periods in which A is not selected.

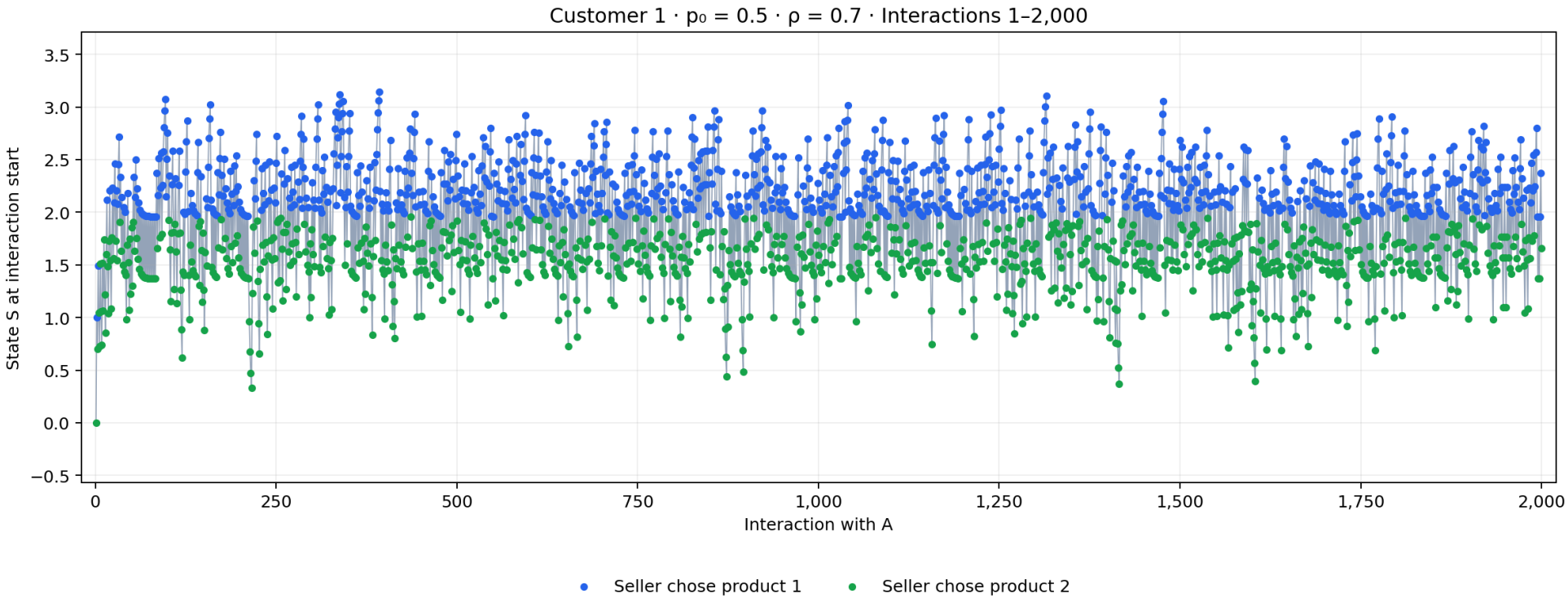
Individual trajectory: $\rho = 0.3$, $p_0 = 0.5$



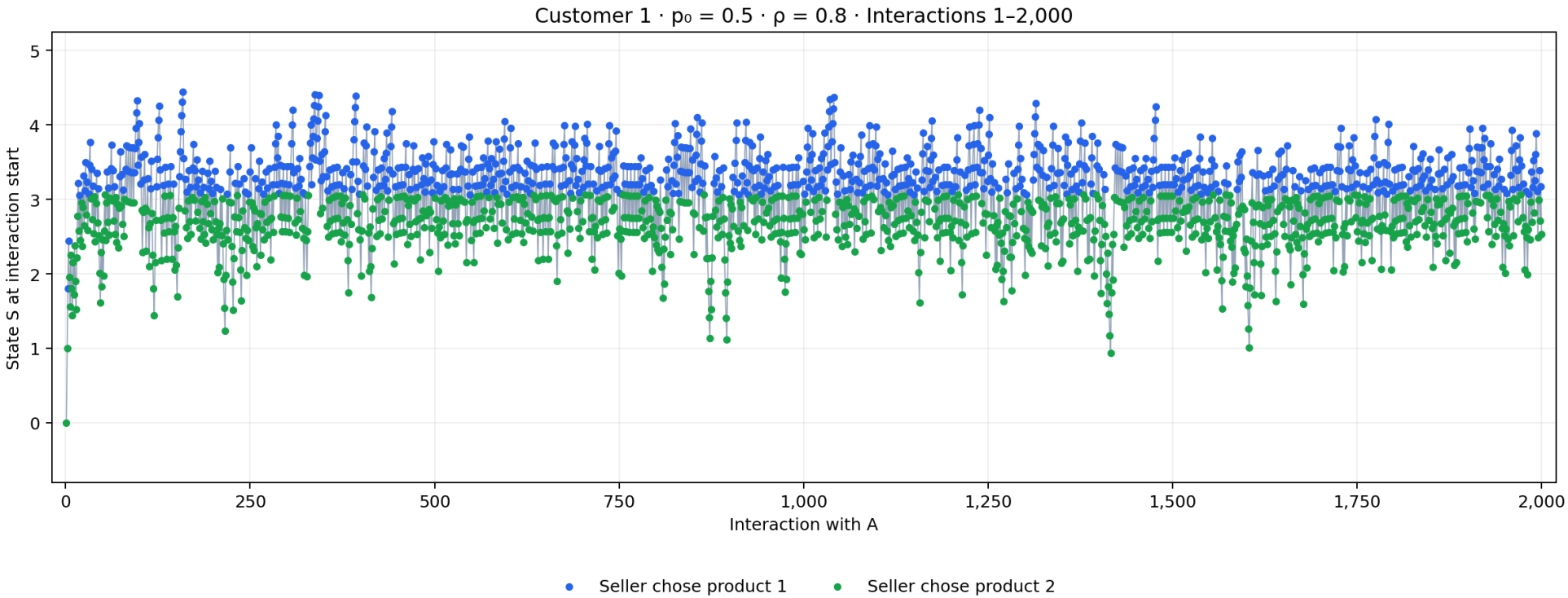
Individual trajectory: $\rho = 0.5$, $p_0 = 0.5$



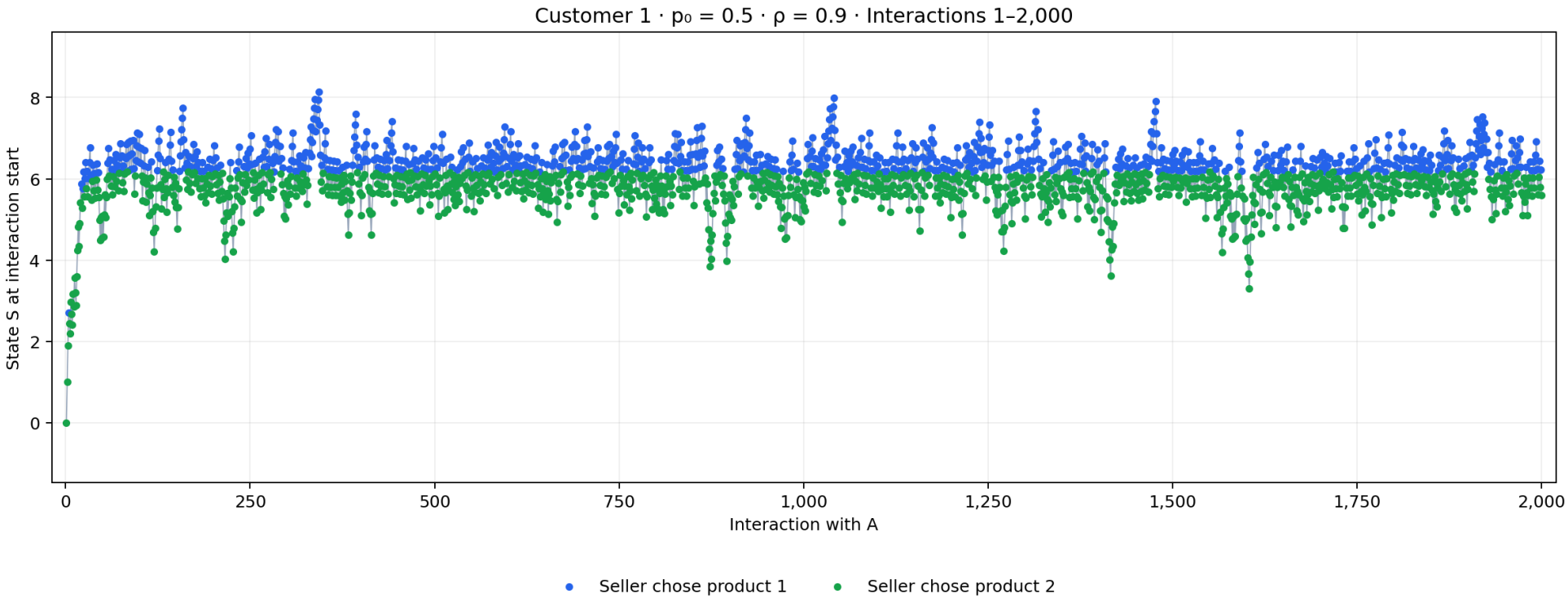
Individual trajectory: $\rho = 0.7$, $p_0 = 0.5$



Individual trajectory: $\rho = 0.8$, $p_0 = 0.5$



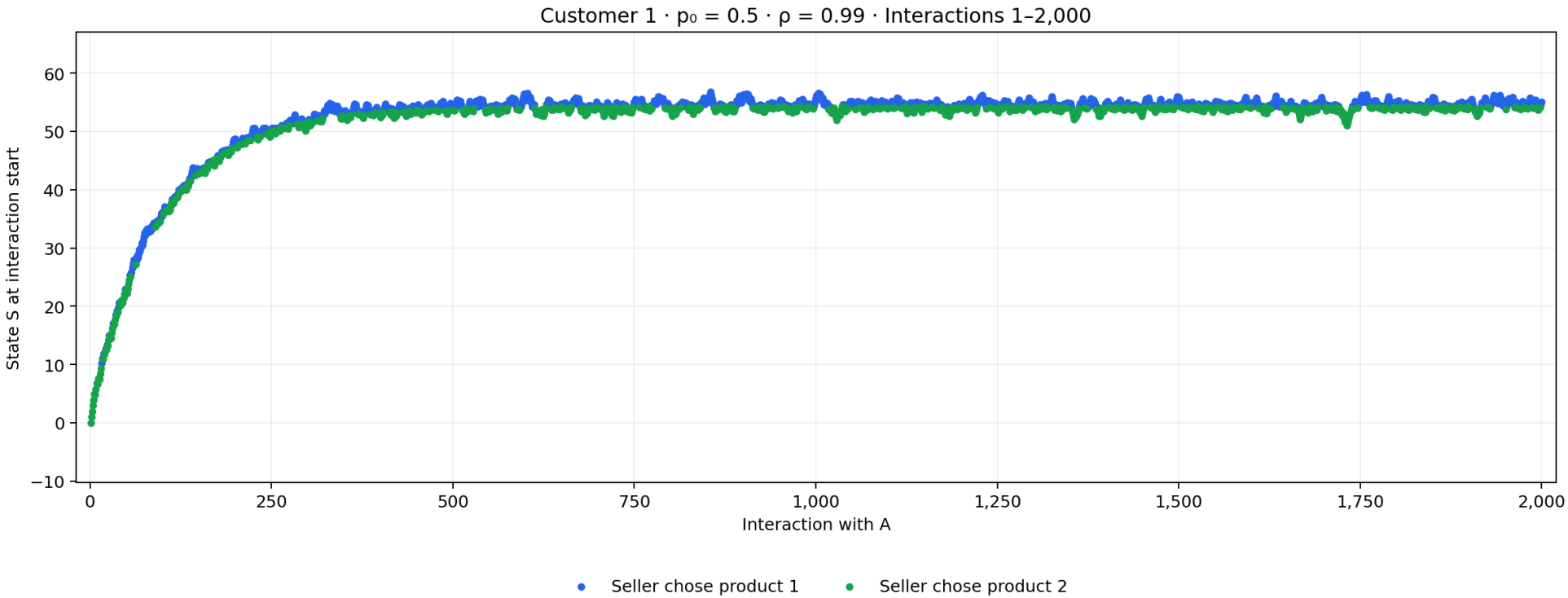
Individual trajectory: $\rho = 0.9$, $p_0 = 0.5$



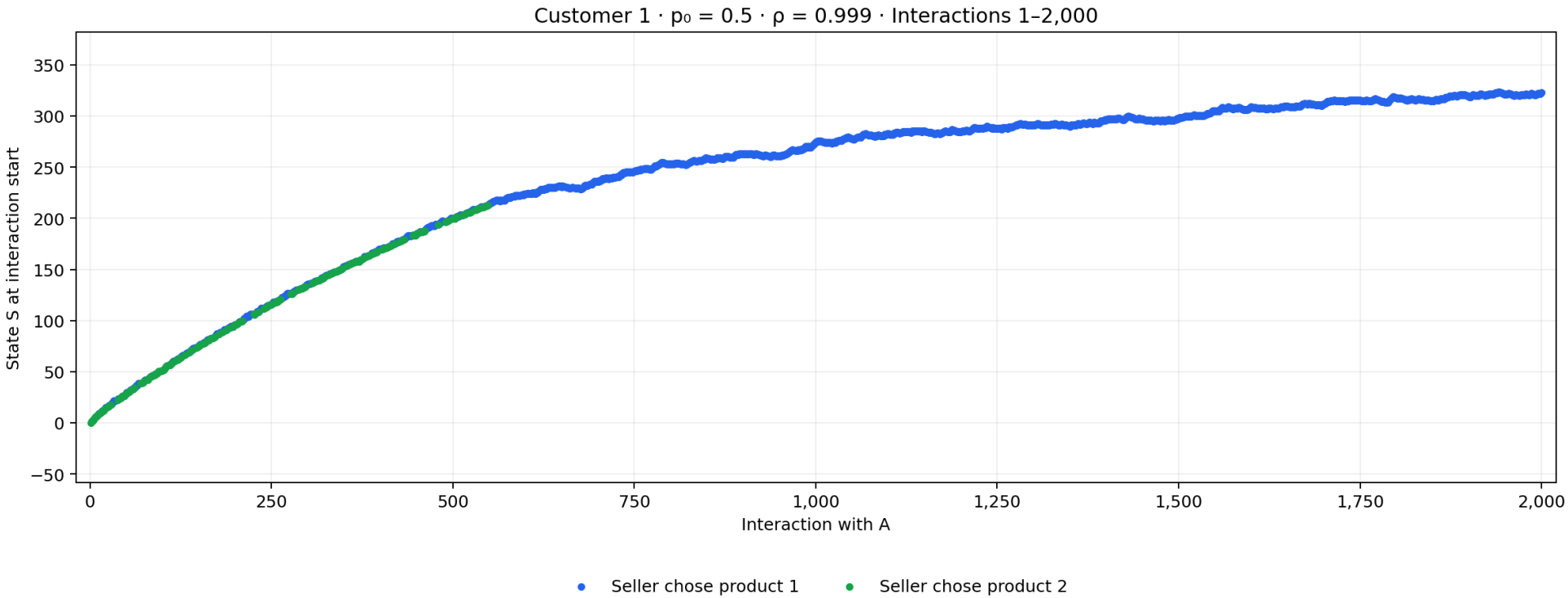
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How are we solving the problem

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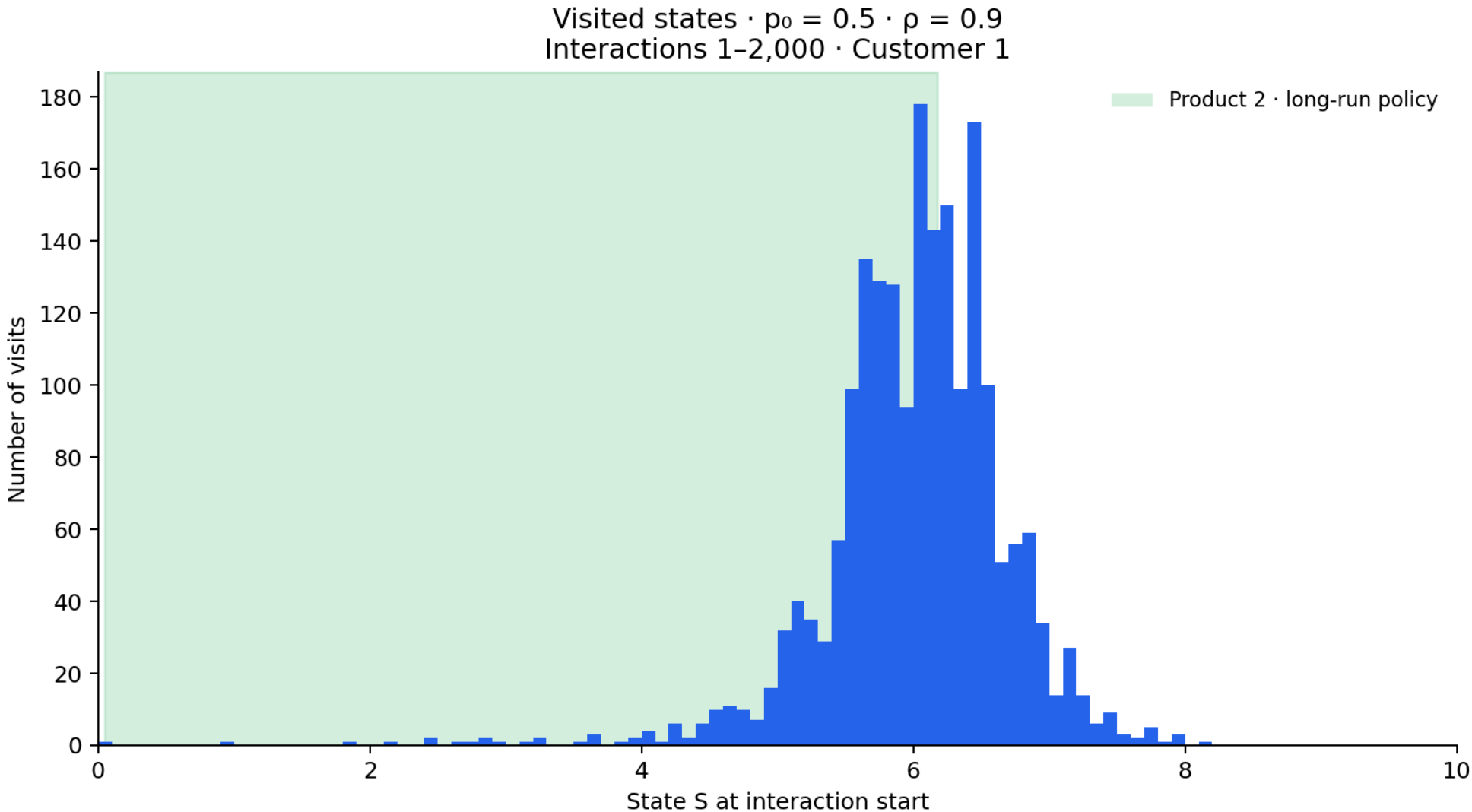
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- Then, in the long run, the state approaches the boundary $S + F = \frac{1}{1-\rho}$.
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- Numerically, we start from the long run boundary and we identify a number K of interactions where the effective information is close to the long run boundary. Then we solve the bellman equation backwards from K to 0.

What we want to do next

- The idea is to first study the set of states where the product 2 is used in the long run and prove that said region is visited infinitey often.

Numerical result $\rho = 0.9$



The idea

- We can study the maximum advantage of product 2 over product 1

$$\mathcal{A}_\rho = \max_{z \in \mathcal{K}_\rho} \{-\Delta c + \Delta p \, G_\rho(z)\}$$

with $G_\rho(z) = \gamma \left[V_\rho\left(T_\rho^s z\right) - V_\rho\left(T_\rho^f z\right) \right]$

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- Then, the intuition is that a sufficiently long sequence of successes and failures can bring the state arbitrarily close to $z^* \in \mathcal{K}_\rho$.Since every such finite sequence has positive probability of occurring, there are repeated opportunities to enter the product-2 region.
- The proof is not ready yet

The idea

- From the numerical experiments we can, in most cases, see thresholds for when the provider decides to use product 2 based on the state
- These thresholds are bounds on the actual visted states

Thresholds and $\mathcal{A}_\rho$ ($p_0 = 0.5$)

ρ	K	Product 2 region	$\mathcal{A}_\rho$	Min S	Max S
0.3	10	$0.000000 \leq S < 0.236868$	0.063933	7.16791×10^{-5}	1.42857
0.5	17	$0.000000 \leq S < 0.890691$	0.433679	0.00655083	1.99935
0.7	33	$0.000000 \leq S < 1.956824$	1.64313	0.138242	3.2993
0.8	52	$0.000000 \leq S < 3.064622$	3.07362	0.591492	4.79443
0.9	110	$0.046307 < S < 6.173594$	2.5117	2.68771	8.5797
0.95	225	$4.156680 < S < 11.965573$	1.78615	8.13693	15.1894
0.99	1146	$41.564085 < S < 54.235753$	0.556712	49.8819	60.8368
0.999	11508	None	-0.170155	322.119	384.329

State S; K is the cut of the boundary solution. None means product 1 throughout. Min and Max S are observed sample extremes in the simulation from 100,001–200,000.

Next steps

- Write the formal proof of the frequent use of the product 2
- Explore ideas on how to find the thresholds analitically